

NEXORA

AI-powered Operational Risk & Supply Intelligence for Semiconductor Fabs

Detect • Predict • Simulate • Act

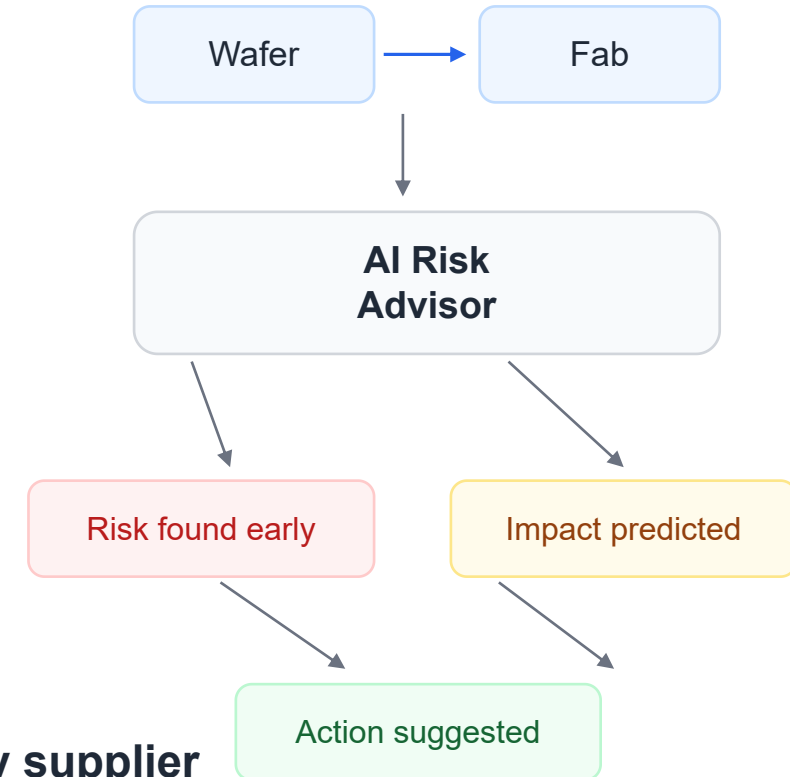
S2 — Fab Bottleneck & Supply Chain Risk Advisor

Team Stack_Trace

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One-line pitch

Nexora helps fabs detect bottlenecks, predict delays, identify supplier risks, and decide the next action.



Live Demo: nexora-7cs9.onrender.com

GitHub: github.com/neel92654/bob-ai-hackathon-Stack_Trace

Why semiconductor fabs need Nexora

The risk is not only one machine or one supplier — it is the chain reaction.

Semiconductor production depends on many machines, materials and suppliers working together.

One overloaded machine can delay downstream steps.

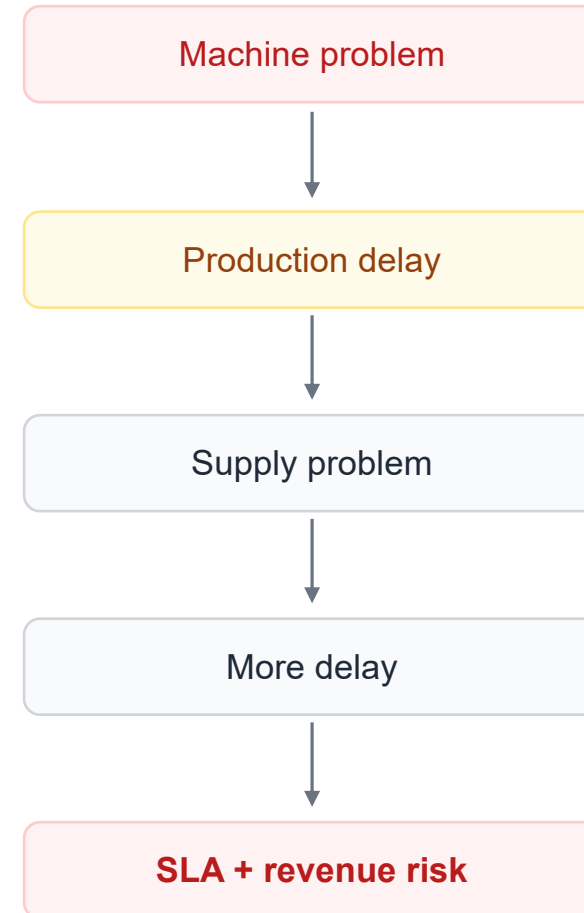
Critical materials may depend on only a few suppliers.

Fab data and supply-chain data often stay disconnected.

Teams discover the full impact after delay has already started.

Real problem

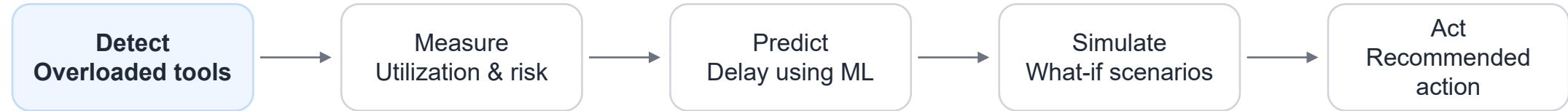
Find risk early → predict impact → recommend the right action.



This chain is what Nexora tries to catch earlier.

Nexora: from detection to decision

A simple decision pipeline for fab operations.



$$\text{Utilization} = \text{WIP} / \text{Capacity} \times 100$$

Example: 65 WIP / 50 Capacity = 130%

130%
critical
utilization

R² 0.954
model fit

Scenario inputs

Supplier embargo
Machine failure
Capacity reduction
Demand surge

**Recommended next
step**

The goal is not only prediction. The goal is decision support.

What makes Nexora different?

Simple explanation, auditable risk logic, and what-if decision support.

1. Physics + AI

Production/queue logic is combined with ML prediction, so the model is not only a black box.

2. Supplier SPoF detection

Dependency, lead time, criticality, geography and alternatives are used to explain supplier risk.

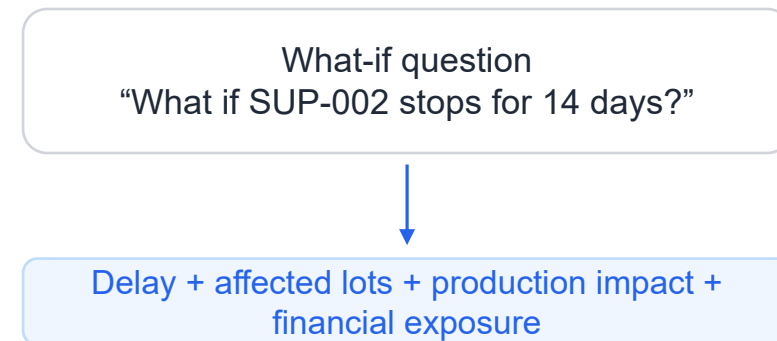
3. What-if simulation

The team can ask: What if this supplier stops for 14 days? Nexora estimates delay and exposure.

Key USP

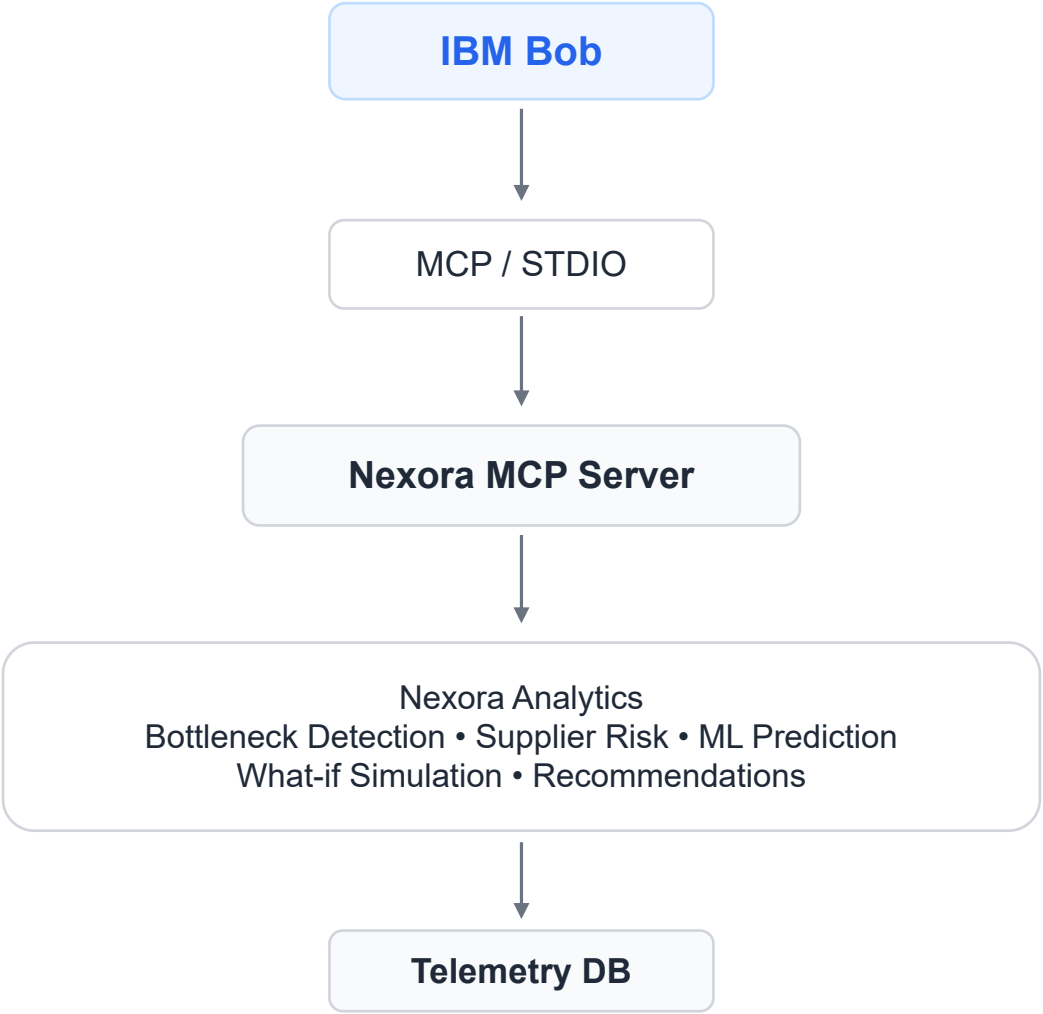
Most systems show what is happening. Nexora helps decide what happens next.

Example risk card



How Nexora works

IBM Bob uses MCP tools to access verified operational data and analysis.



Technology

Frontend	React + Vite
Backend	Python + Flask
ML	Random Forest
AI	IBM Bob + MCP
DB / Tests	SQLite / Pytest

Why this matters

Bob is not guessing from static text. It calls Nexora tools and receives grounded operational outputs.

Working prototype and results

The slide judges should remember: built, tested, and measurable.

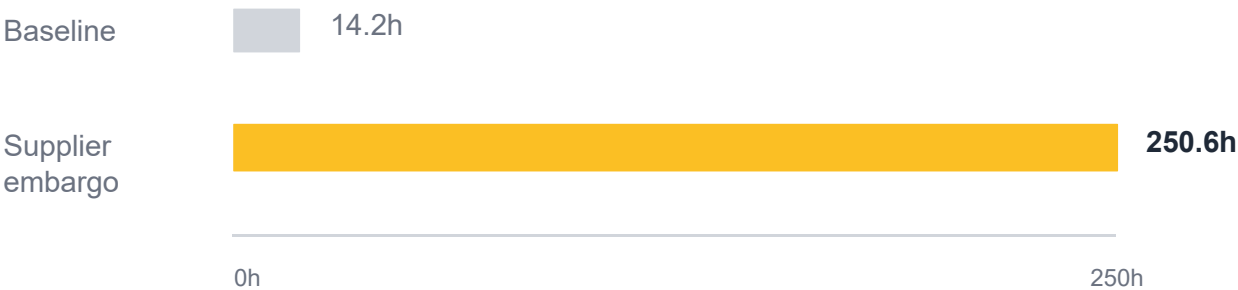
**130%
CVD-03
utilization**

**R² 0.954
ML model fit**

**+236.4h
extra delay**

**52 / 52
tests passed**

What-if simulation: baseline vs supplier embargo



Prototype output snapshot

Tool	Result
Bottleneck detector	CVD-03 critical
Delay predictor	MAE 2.81h
Scenario simulator	+236.4h delay
Risk advisor	\$2.4M exposure

This proves the prototype is more than a UI mockup: it detects, predicts, simulates and returns an action-oriented result.

Making semiconductor operations more resilient

Impact today, roadmap for real fab integration tomorrow.

Expected impact

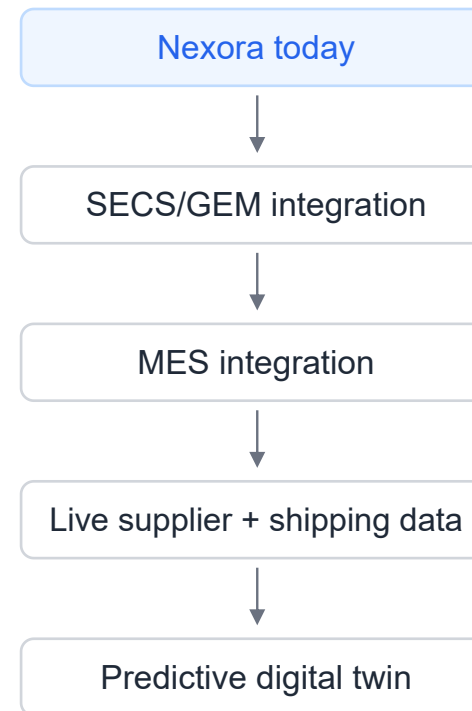
- Faster bottleneck identification before delays become bigger.
- Lower SLA and downtime exposure through earlier decisions.
- Explainable and auditable risk predictions.
- Better supplier resilience by detecting critical dependency early.

Nexora predicts operational risk before it becomes production loss.

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Future roadmap



Thank you — Team Stack_Trace